

1605.05775: equation evidence

Display equations

Identifier	Page	Conf.	LaTeX source	Rendered	Image
1605.05775_ EQ0014 (17)	7	■ 0.927	$\begin{aligned} & W_{s_1 s_2 \cdots s_N}^{\ell} = \\ & \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}} \end{aligned}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} U_{s_1}^{\alpha_1} \cdots U_{\alpha_{i-1} s_i}^{\alpha_i} C_{\alpha_i \alpha_{i+1}}^{\ell} V_{s_{i+1} \alpha_{i+2}}^{\alpha_{i+1}} \cdots V_{s_N}^{\alpha_{N-1}}$
1605.05775_ EQ0026 (B4)	10	■ 0.988	$\phi^{\{s_j\}} \left(x_j \right) = \sqrt{\binom{d-1}{s_j-1}} \left(\cos \left(\frac{\pi}{2} x_j \right) \right)^{d-s_j} \left(\sin \left(\frac{\pi}{2} x_j \right) \right)^{s_j-1}$	$\phi^{s_j}(x_j) = \sqrt{\binom{d-1}{s_j-1}} \left(\cos \left(\frac{\pi}{2} x_j \right) \right)^{d-s_j} \left(\sin \left(\frac{\pi}{2} x_j \right) \right)^{s_j-1}$	$\phi^{s_j}(x_j) = \sqrt{\binom{d-1}{s_j-1}} \left(\cos \left(\frac{\pi}{2} x_j \right) \right)^{d-s_j} \left(\sin \left(\frac{\pi}{2} x_j \right) \right)^{s_j-1}$
1605.05775_ EQ0013 (15)	7	■ 0.989	$\begin{aligned} & \phi(0) = [1, 0] \\ & \phi(1) = [0, 1] \end{aligned}$	$\begin{aligned} \phi(0) &= [1, 0] \\ \phi(1) &= [0, 1]. \end{aligned}$	$\begin{aligned} \phi(0) &= [1, 0] \\ \phi(1) &= [0, 1] \end{aligned}$
1605.05775_ EQ0025 (B3)	10	■ 0.990	$\begin{aligned} & \left(\cos^2(\theta_j) + \sin^2(\theta_j) \right)^{d-1} = 1 \\ & = \sum_{p=0}^{d-1} \binom{d-1}{p} (\cos \theta_j)^{2(d-1-p)} (\sin \theta_j)^{2p} \end{aligned}$	$\begin{aligned} \left(\cos^2(\theta_j) + \sin^2(\theta_j) \right)^{d-1} &= 1 \\ &= \sum_{p=0}^{d-1} \binom{d-1}{p} (\cos \theta_j)^{2(d-1-p)} (\sin \theta_j)^{2p} \end{aligned}$	$\begin{aligned} (\cos^2(\theta_j) + \sin^2(\theta_j))^{d-1} &= 1 \\ &= \sum_{p=0}^{d-1} \binom{d-1}{p} (\cos \theta_j)^{2(d-1-p)} (\sin \theta_j)^{2p} \end{aligned}$
1605.05775_ EQ0005 (5)	3	■ 0.995	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$	$W_{s_1 s_2 \cdots s_N}^{\ell} = \sum_{\{\alpha\}} A_{s_1}^{\alpha_1} A_{s_2}^{\alpha_1 \alpha_2} \cdots A_{s_j}^{\ell; \alpha_j \alpha_{j+1}} \cdots A_{s_N}^{\alpha_{N-1}}$
1605.05775_ EQ0008 (9)	4	■ 0.999	$\begin{aligned} & \Delta B^{\ell} \stackrel{\text{def}}{=} - \frac{\partial C}{\partial B^{\ell}} \\ & = \sum_{n=1}^{N_T} \sum_{\ell'} \left(\delta_{L_n}^{\ell'} - f^{\ell'}(\mathbf{x}_n) \right) \frac{\partial f^{\ell'}(\mathbf{x}_n)}{\partial B^{\ell}} \\ & = \sum_{n=1}^{N_T} \left(\delta_{L_n}^{\ell} - f^{\ell}(\mathbf{x}_n) \right) \tilde{\Phi}_n \end{aligned}$	$\begin{aligned} \Delta B^{\ell} &\stackrel{\text{def}}{=} - \frac{\partial C}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \sum_{\ell'} \left(\delta_{L_n}^{\ell'} - f^{\ell'}(\mathbf{x}_n) \right) \frac{\partial f^{\ell'}(\mathbf{x}_n)}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \left(\delta_{L_n}^{\ell} - f^{\ell}(\mathbf{x}_n) \right) \tilde{\Phi}_n \end{aligned}$	$\begin{aligned} \Delta B^{\ell} &\stackrel{\text{def}}{=} - \frac{\partial C}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \sum_{\ell'} \left(\delta_{L_n}^{\ell'} - f^{\ell'}(\mathbf{x}_n) \right) \frac{\partial f^{\ell'}(\mathbf{x}_n)}{\partial B^{\ell}} \\ &= \sum_{n=1}^{N_T} \left(\delta_{L_n}^{\ell} - f^{\ell}(\mathbf{x}_n) \right) \tilde{\Phi}_n \end{aligned}$
1605.05775_ EQ0009 (11)	4	■ 0.999	$B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} = \sum_{\alpha'_j \alpha_j} U_{s_j \alpha'_j}^{\alpha_{j-1}} S^{\alpha'_j}_{\alpha_j} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}},$	$B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} = \sum_{\alpha'_j \alpha_j} U_{s_j \alpha'_j}^{\alpha_{j-1}} S^{\alpha'_j}_{\alpha_j} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}},$	$B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} = \sum_{\alpha'_j \alpha_j} U_{s_j \alpha'_j}^{\alpha_{j-1}} S^{\alpha'_j}_{\alpha_j} V_{s_{j+1}}^{\alpha_j \ell \alpha_{j+1}},$

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1605.05775_ EQ0001 (1)	1	■ 1.000	$f(\mathbf{x})=W \cdot \Phi(\mathbf{x})$.	$f(\mathbf{x}) = W \cdot \Phi(\mathbf{x})$.	$f(\mathbf{x}) = W \cdot \Phi(\mathbf{x})$.
1605.05775_ EQ0002 (2)	2	■ 1.000	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x})=\phi^{s_1}(\mathbf{x}_1) \otimes \phi^{s_2}(\mathbf{x}_2) \otimes \cdots \phi^{s_N}(\mathbf{x}_N)$.	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x}) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$.	$\Phi^{s_1 s_2 \cdots s_N}(\mathbf{x}) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$.
1605.05775_ EQ0003 (3)	2	■ 1.000	$\phi^{s_j}(\mathbf{x}_j)=\cos(\frac{\pi}{2}x_j), \sin(\frac{\pi}{2}x_j)$	$\phi^{s_j}(x_j) = \left[\cos\left(\frac{\pi}{2}x_j\right), \sin\left(\frac{\pi}{2}x_j\right) \right]$	$\phi^{s_j}(x_j) = \left[\cos\left(\frac{\pi}{2}x_j\right), \sin\left(\frac{\pi}{2}x_j\right) \right]$
1605.05775_ EQ0004 (4)	2	■ 1.000	$f^{\ell}(\mathbf{x})=W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$	$f^{\ell}(\mathbf{x}) = W^{\ell} \cdot \Phi(\mathbf{x})$
1605.05775_ EQ0006 (6)	3	■ 1.000	$C=\frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} \left(f^{\ell}(\mathbf{x}_n)-\delta_{L_n}^{\ell}\right)^2$	$C = \frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} \left(f^{\ell}(\mathbf{x}_n) - \delta_{L_n}^{\ell}\right)^2$	$C = \frac{1}{2} \sum_{n=1}^{N_T} \sum_{\ell} \left(f^{\ell}(\mathbf{x}_n) - \delta_{L_n}^{\ell}\right)^2$
1605.05775_ EQ0007 (7)	4	■ 1.000	$f^{\ell}(\mathbf{x}_n)=\sum_{\alpha_j} \alpha_{j+1} B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} \left(\tilde{\Phi}_n\right)^{s_j s_{j+1}}$	$f^{\ell}(\mathbf{x}_n) = \sum_{\alpha_{j-1} \alpha_{j+1}} \sum_{s_j s_{j+1}} B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} \left(\tilde{\Phi}_n\right)^{s_j s_{j+1}}$	$f^{\ell}(\mathbf{x}_n) = \sum_{\alpha_{j-1} \alpha_{j+1}} \sum_{s_j s_{j+1}} B_{s_j s_{j+1}}^{\alpha_{j-1} \ell \alpha_{j+1}} \left(\tilde{\Phi}_n\right)^{s_j s_{j+1}}$
1605.05775_ EQ0010 (12)	5	■ 1.000	$\Phi(\mathbf{x}_1, \mathbf{x}_2)=\phi^{s_1}(\mathbf{x}_1) \otimes \phi^{s_2}(\mathbf{x}_2)$	$\Phi(x_1, x_2) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2)$	$\Phi(x_1, x_2) = \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2)$
1605.05775_ EQ0011 (13)	7	■ 1.000	$f(\mathbf{x})=\sum_s W_{s_1 s_2 \cdots s_N} \phi^{s_1}(\mathbf{x}_1) \otimes \phi^{s_2}(\mathbf{x}_2) \otimes \cdots \phi^{s_N}(\mathbf{x}_N)$.	$f(\mathbf{x}) = \sum_{\{s\}} W_{s_1 s_2 \cdots s_N} \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$.	$f(\mathbf{x}) = \sum_{\{s\}} W_{s_1 s_2 \cdots s_N} \phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$.
1605.05775_ EQ0012 (14)	7	■ 1.000	$\phi^{s_1}(\mathbf{x}_1) \otimes \phi^{s_2}(\mathbf{x}_2) \otimes \cdots \phi^{s_N}(\mathbf{x}_N)$	$\phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$	$\phi^{s_1}(x_1) \otimes \phi^{s_2}(x_2) \otimes \cdots \phi^{s_N}(x_N)$
1605.05775_ EQ0015 (18)	8	■ 1.000	$\sum_{\ell} \int_{\mathbf{x}} \left f^{\ell}(\mathbf{x})\right ^2 d \mu(\mathbf{x})=1$	$\sum_{\ell} \int_{\mathbf{x}} \left f^{\ell}(\mathbf{x})\right ^2 d \mu(x) = 1$	$\sum_{\ell} \int_{\mathbf{x}} \left f^{\ell}(\mathbf{x})\right ^2 d \mu(x) = 1$
1605.05775_ EQ0016 (19)	8	■ 1.000	$\sum_{\ell} \sum_{s_1, s_2, \ldots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1$.	$\sum_{\ell} \sum_{s_1, s_2, \ldots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1$.	$\sum_{\ell} \sum_{s_1, s_2, \ldots, s_N} \bar{W}_{s_1 s_2 \cdots s_N}^{\ell} W_{s_1 s_2 \cdots s_N}^{\ell} = 1$.
1605.05775_ EQ0017 (20)	8	■ 1.000	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d \mu(x)=\delta_{ss'}$	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d \mu(x) = \delta_{ss'}$	$\int_x \bar{\phi}^s(x) \phi^{s'}(x) d \mu(x) = \delta_{ss'}$,
1605.05775_ EQ0018 (21)	8	■ 1.000	$\sum_s \left \phi^s(\mathbf{x})\right ^2=1$	$\sum_s \left \phi^s(x)\right ^2 = 1$	$\sum_s \left \phi^s(x)\right ^2 = 1$
1605.05775_ EQ0019 (22)	8	■ 1.000	$\phi(x)=[\cos(\pi x), \sin(\pi x)]$.	$\phi(x) = [\cos(\pi x), \sin(\pi x)]$.	$\phi(x) = \left[\cos(\pi x), \sin(\pi x) \right]$.
1605.05775_ EQ0020 (23)	8	■ 1.000	$\phi(x)=\left[e^{i(3 \pi / 2) x} \cos \left(\frac{\pi}{2} x\right), e^{-i(3 \pi / 2) x} \sin \left(\frac{\pi}{2} x\right)\right]$.	$\phi(x) = \left[e^{i(3 \pi / 2) x} \cos \left(\frac{\pi}{2} x\right), e^{-i(3 \pi / 2) x} \sin \left(\frac{\pi}{2} x\right)\right]$.	$\phi(x) = \left[e^{i(3 \pi / 2) x} \cos \left(\frac{\pi}{2} x\right), e^{-i(3 \pi / 2) x} \sin \left(\frac{\pi}{2} x\right)\right]$.
1605.05775_ EQ0021 (24)	9	■ 1.000	$C=-\sum_{n=1}^{N_s} \log \left f^{L_n}(\mathbf{x}_n)\right ^2$	$C = -\sum_{n=1}^{N_s} \log \left f^{L_n}(\mathbf{x}_n)\right ^2$	$C = -\sum_{n=1}^{N_s} \log \left f^{L_n}(\mathbf{x}_n)\right ^2$

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1605.05775_ E00022 (25)	9	■ 1.000	$D_{\mathrm{KL}}=\sum_{\ell} \int_{\mathbf{x}} p(\ell, \mathbf{x}) \log \left(\frac{p(\ell, \mathbf{x})}{\tilde{p}(\ell, \mathbf{x})} \right)$	$D_{\mathrm{KL}} = \sum_{\ell} \int_{\mathbf{x}} p(\ell, \mathbf{x}) \log \left(\frac{p(\ell, \mathbf{x})}{\tilde{p}(\ell, \mathbf{x})} \right)$	$D_{\mathrm{KL}} = \sum_{\ell} \int_{\mathbf{x}} p(\ell, \mathbf{x}) \log \left(\frac{p(\ell, \mathbf{x})}{\tilde{p}(\ell, \mathbf{x})} \right)$
1605.05775_ E00023 (B1)	10	■ 1.000	$\phi^{s_j}(x_j) = \left[\cos \left(\frac{\pi}{2} x_j \right), \sin \left(\frac{\pi}{2} x_j \right) \right]$	$\phi^{s_j}(x_j) = \left[\cos \left(\frac{\pi}{2} x_j \right), \sin \left(\frac{\pi}{2} x_j \right) \right]$	$\phi^{s_j}(x_j) = \left[\cos \left(\frac{\pi}{2} x_j \right), \sin \left(\frac{\pi}{2} x_j \right) \right]$
1605.05775_ E00024 (B2)	10	■ 1.000	$\left(\cos^2(\theta_j) + \sin^2(\theta_j) \right)^{d-1} = 1$	$\left(\cos^2(\theta_j) + \sin^2(\theta_j) \right)^{d-1} = 1$	$(\cos^2(\theta_j) + \sin^2(\theta_j))^{d-1} = 1$